

# Problem of the Week 11

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## Motivation

This week's Problem of the Week is focused on comparing and contrasting the neural network architectures we've seen during the past few weeks.

## Completing this assignment

Fill in the cell as indicated below. When you are happy with your response, produce a clean rendition of your notebook. Then, create a PDF of your notebook and upload both documents (PDF and ipynb file) to Canvas.

## Problem 11.1 - Overview of Neural Networks

One of your friends comes to you and asks what you've been learning in your CS 171 class. You tell them: "neural networks". They look at you perplexed so you elaborate: "you know, feed-forward neural networks, convolutional neural networks, and recurrent neural networks". They look even more perplexed. Start by explaining to them what a neural network is in two sentences.

A neural network is kinda like a brain. It takes in an input, like our senses hearing words and seeing body language, and spits out an output, like our own speech and body language. There's a bunch of magic happening in the middle that nobody really knows the exact details of but just like how we learn, neural networks learn what inputs correspond to what outputs and tunes itself to achieve the best result it can.

## Problem 11.2 - Model Purposes

For each of the NNs listed above, explain what type of data they are designed to model.

Feed-forward NN: Lists (vectors) of numbers.

Convolutional NN: Data that is arranged in 2 dimensional grids, like images.

Recurrent NN: Data that is time-based or otherwise sequential.

## Problem 11.3 - Model Types

Your friend is starting to understand a little bit better. Now, in 2-3 sentences, explain how a CNN differs from an RNN. How do each of these differ from a "regular" or "feed-forward" neural network?

A CNN aims to learn spatially from the data, trying to figure out how features influence each other in a 2D form. Meanwhile an RNN aims to learn from how previous states compare to current states. A regular NN just tries to figure out the best combination of weights in a list and are far less complex.